OLS and Linear Algebra

A visual, beginner-friendly guide to matrix derivatives, rank, and invertibility

Teaching notes for an introductory machine-learning class

The aim. This guide starts with scalar equations and builds towards matrix notation. It explains what each matrix product means component-wise, what a derivative with respect to a vector actually represents, how the OLS normal equations arise, and why $X^\top X$ is invertible exactly when the columns of X are linearly independent.

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1 The objects: scalars, vectors, and matrices

1.1 Scalars

A *scalar* is one number, for example $a = 3$. A scalar has a value but no row or column dimension.

1.2 Vectors

A *column vector* is a list of numbers arranged vertically:

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} \in \mathbb{R}^3.$$

Here $\beta_0, \beta_1, \beta_2$ are three separate scalar parameters. Writing them as one vector is a compact way to keep track of them together.

The transpose changes a column into a row:

$$\boldsymbol{\beta}^{\mathsf{T}} = \begin{bmatrix} \beta_0 & \beta_1 & \beta_2 \end{bmatrix}.$$

Transposition does not change the entries; it changes their arrangement.

1.3 Matrices

A matrix is a rectangular array. For example, a design matrix with n observations and p coefficients is

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \in \mathbb{R}^{n \times p}.$$

The entry x_{ij} is in row i, column j. In regression, row i contains the features for observation i; column j contains feature j across all observations.

Dimensions matter. If A has size $m \times n$ and B has size $r \times s$, then AB is defined only when $n = r$. The product then has size $m \times s$:

$$\underbrace{(m \times n)}_{A}\underbrace{(n \times s)}_{B} = \underbrace{(m \times s)}_{AB}.$$

The inner dimensions must match; the outer dimensions give the answer's shape.

1.4 The regression objects and their dimensions

$$\underbrace{X}_{n \times p} \quad \underbrace{\boldsymbol{\beta}}_{p \times 1} \quad \underbrace{\boldsymbol{y}}_{n \times 1}.$$

Object	Shape	Meaning
X	$n \times p$	Design matrix / features
$\boldsymbol{\beta}$	$p \times 1$	Unknown coefficients
$\boldsymbol{y}$	$n \times 1$	Observed outcomes
$X\boldsymbol{\beta}$	$n \times 1$	Predicted outcomes
$\boldsymbol{e} = \boldsymbol{y} - X\boldsymbol{\beta}$	$n \times 1$	Residuals
$\boldsymbol{e}^\top \boldsymbol{e}$	1×1	Sum of squared residuals, a scalar
$X^\top X$	$p \times p$	Gram matrix of feature columns
$X^\top \boldsymbol{y}$	$p \times 1$	Right-hand side of the normal equations

2 Matrix multiplication, unpacked

2.1 Why $X\beta$ gives all predictions

Take the small design matrix

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \qquad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}.$$

Then

$$X\boldsymbol{\beta} = \begin{bmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} \beta_0 + 2\beta_1 \\ \beta_0 + 4\beta_1 \\ \beta_0 + 5\beta_1 \end{bmatrix}.$$

Each output entry is the dot product of one row of X with the coefficient vector:

$$(X\boldsymbol{\beta})_i = \sum_{j=1}^{p} x_{ij}\beta_j.$$

If the first column consists of ones, its coefficient is an intercept.

2.2 Dot products and squared length

For two vectors of the same length,

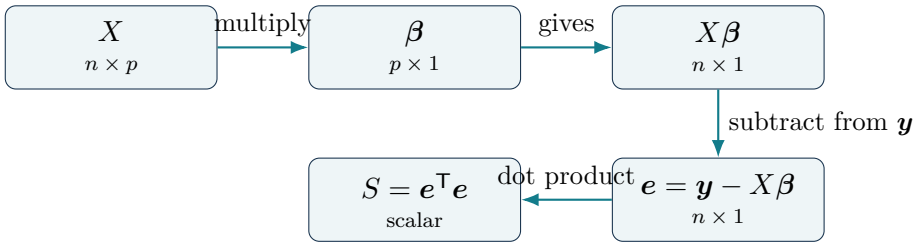
$$\boldsymbol{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}, \quad \boldsymbol{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, \qquad \boldsymbol{a}^\top \boldsymbol{b} = a_1 b_1 + a_2 b_2 + a_3 b_3.$$

The result is a scalar. In particular,

$$\boldsymbol{a}^\top \boldsymbol{a} = a_1^2 + a_2^2 + a_3^2 = \|\boldsymbol{a}\|^2 \geq 0.$$

This fact will explain why $X^\top X$ is positive semidefinite.

2.3 A visual map of the OLS dimensions



3 Derivatives: the scalar rules first

3.1 Derivative of a scalar function

For a function of one scalar variable, $f(\beta)$, the derivative $f'(\beta)$ measures the local rate of change of f when β changes.

Useful rules:

$$\frac{\mathrm{d}}{\mathrm{d}\beta}c = 0 \qquad (c \text{ is constant}),$$
$$\frac{\mathrm{d}}{\mathrm{d}\beta}(a\beta) = a,$$
$$\frac{\mathrm{d}}{\mathrm{d}\beta}\beta^2 = 2\beta,$$
$$\frac{\mathrm{d}}{\mathrm{d}\beta}f(g(\beta)) = f'(g(\beta))g'(\beta) \quad \text{(chain rule)}.$$

For example, with x, y fixed,

$$f(\beta) = (y - x\beta)^2 \quad \Longrightarrow \quad f'(\beta) = 2(y - x\beta)(-x).$$

The factor $-x$ appears because $\frac{\mathrm{d}}{\mathrm{d}\beta}(y - x\beta) = -x$.

3.2 Several parameters: partial derivatives

Suppose a scalar function depends on two parameters:

$$f(\beta_1, \beta_2) = (y - \beta_1 - 2\beta_2)^2.$$

To find the effect of β_1, hold β_2 fixed. To find the effect of β_2, hold β_1 fixed:

$$\frac{\partial f}{\partial \beta_1} = -2(y - \beta_1 - 2\beta_2), \qquad \frac{\partial f}{\partial \beta_2} = -4(y - \beta_1 - 2\beta_2).$$

The *gradient* stacks these partial derivatives:

$$\nabla_{\boldsymbol{\beta}} f = \begin{bmatrix} \frac{\partial f}{\partial \beta_1} \\ \frac{\partial f}{\partial \beta_2} \end{bmatrix}.$$

For p parameters,

$$\nabla_{\boldsymbol{\beta}} f = \begin{bmatrix} \partial f/\partial \beta_1 \\ \vdots \\ \partial f/\partial \beta_p \end{bmatrix} \in \mathbb{R}^p.$$

Interpretation: $\partial f/\partial \beta_j$ is one ordinary scalar derivative, taken while the other parameters are held fixed. The gradient is simply the list of all these derivatives arranged as a column vector.

3.3 What is a derivative *with respect to a vector*?

The phrase is incomplete until we know the type of the output. A vector input does not force every derivative to be a vector.

Function	Output type	Derivative (using the convention in this guide)
$f(\boldsymbol{\beta})$	scalar	Gradient, a $p \times 1$ vector
$\boldsymbol{g}(\boldsymbol{\beta}) \in \mathbb{R}^m$	vector	Jacobian, an $m \times p$ matrix
$A(\boldsymbol{\beta}) \in \mathbb{R}^{m \times n}$	matrix	Differentiate each entry; the full collection has three indices (a tensor)

If $\boldsymbol{g}(\boldsymbol{\beta}) = [g_1, \ldots, g_m]^\mathsf{T}$, the Jacobian is

$$J_{\boldsymbol{g}}(\boldsymbol{\beta}) = \begin{bmatrix} \frac{\partial g_1}{\partial \beta_1} & \cdots & \frac{\partial g_1}{\partial \beta_p} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_m}{\partial \beta_1} & \cdots & \frac{\partial g_m}{\partial \beta_p} \end{bmatrix} \in \mathbb{R}^{m \times p}.$$

For the linear map $\boldsymbol{g}(\boldsymbol{\beta}) = X\boldsymbol{\beta}$, component (i, j) of the Jacobian is

$$\frac{\partial (X\boldsymbol{\beta})_i}{\partial \beta_j} = x_{ij}, \qquad \text{so} \qquad J_{\boldsymbol{g}}(\boldsymbol{\beta}) = X.$$

In OLS, the objective $S(\boldsymbol{\beta})$ is a *scalar*. Therefore its derivative with respect to the coefficient vector is a gradient vector. We are not taking the derivative of the design matrix X with respect to $\boldsymbol{\beta}$; X is treated as fixed data.

4 OLS first in scalar form

For observation i, the prediction is

$$\hat{y}_i = \sum_{j=1}^{p} x_{ij}\beta_j, \qquad e_i = y_i - \hat{y}_i.$$

OLS chooses coefficients to minimise the sum of squared residuals:

$$S(\beta_1, \ldots, \beta_p) = \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{p} x_{ij}\beta_j \right)^2.$$

Why square the residuals? Positive and negative errors should not cancel, and large errors are penalised more heavily.

Differentiate with respect to one coefficient, β_k. By the chain rule,

$$\frac{\partial S}{\partial \beta_k} = -2 \sum_{i=1}^{n} x_{ik} \left(y_i - \sum_{j=1}^{p} x_{ij}\beta_j \right).$$

At an unconstrained differentiable minimum, each partial derivative is zero:

$$\sum_{i=1}^{n} x_{ik} \left(y_i - \sum_{j=1}^{p} x_{ij}\beta_j \right) = 0, \qquad k = 1, \ldots, p.$$

There are p equations, one for each coefficient.

The scalar derivation says that each feature column has zero dot product with the residual vector at the solution. In matrix notation, these p equations become $X^\mathsf{T}(\boldsymbol{y} - X\hat{\boldsymbol{\beta}}) = \mathbf{0}$.

5 OLS in matrix form, with every step explained

5.1 Step 1: stack the predictions and residuals

$$\hat{\boldsymbol{y}} = X\boldsymbol{\beta}, \qquad \boldsymbol{e} = \boldsymbol{y} - X\boldsymbol{\beta}.$$

This is exactly the scalar prediction equation applied to every row of X.

5.2 Step 2: write the sum of squares as a product

$$S(\boldsymbol{\beta}) = \boldsymbol{e}^\mathsf{T}\boldsymbol{e} = (\boldsymbol{y} - X\boldsymbol{\beta})^\mathsf{T}(\boldsymbol{y} - X\boldsymbol{\beta}).$$

To see why this is the sum of squares, write

$$\boldsymbol{e}^\mathsf{T}\boldsymbol{e} = \begin{bmatrix} e_1 & e_2 & \cdots & e_n \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix} = \sum_{i=1}^{n} e_i^2.$$

The dimensions are $(1 \times n)(n \times 1) = (1 \times 1)$: the objective is a scalar.

5.3 Step 3: expand using transpose rules

Two rules are used:

$$(A - B)^\mathsf{T} = A^\mathsf{T} - B^\mathsf{T}, \qquad (AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}.$$

Hence

$$\begin{aligned} S(\boldsymbol{\beta}) &= (\boldsymbol{y} - X\boldsymbol{\beta})^\mathsf{T}(\boldsymbol{y} - X\boldsymbol{\beta}) \\ &= (\boldsymbol{y}^\mathsf{T} - \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T})(\boldsymbol{y} - X\boldsymbol{\beta}) \\ &= \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - \boldsymbol{y}^\mathsf{T}X\boldsymbol{\beta} - \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}\boldsymbol{y} + \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}X\boldsymbol{\beta}. \end{aligned}$$

The middle two terms are scalars and are transposes of one another:

$$\boldsymbol{y}^\mathsf{T}X\boldsymbol{\beta} = (\boldsymbol{y}^\mathsf{T}X\boldsymbol{\beta})^\mathsf{T} = \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}\boldsymbol{y}.$$

Therefore,

$$\boxed{S(\boldsymbol{\beta}) = \boldsymbol{y}^\mathsf{T}\boldsymbol{y} - 2\boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}\boldsymbol{y} + \boldsymbol{\beta}^\mathsf{T}X^\mathsf{T}X\boldsymbol{\beta}.}$$

5.4 Step 4: derivative rules for vectors and matrices

Treat X and $\boldsymbol{y}$ as constants. The rules needed are

$$\begin{aligned} \nabla_{\boldsymbol{\beta}} c &= \mathbf{0}, \\ \nabla_{\boldsymbol{\beta}}(\boldsymbol{a}^\mathsf{T}\boldsymbol{\beta}) &= \boldsymbol{a}, \\ \nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\mathsf{T}\boldsymbol{a}) &= \boldsymbol{a}, \\ \nabla_{\boldsymbol{\beta}}(\boldsymbol{\beta}^\mathsf{T}A\boldsymbol{\beta}) &= (A + A^\mathsf{T})\boldsymbol{\beta}. \end{aligned}$$

The last rule can be seen component-wise. Write

$$q(\boldsymbol{\beta}) = \boldsymbol{\beta}^\mathsf{T}A\boldsymbol{\beta} = \sum_{i=1}^{p}\sum_{j=1}^{p} \beta_i a_{ij} \beta_j.$$

Differentiating with respect to β_k picks up terms where β_k appears on the left and terms where it appears on the right:

$$\frac{\partial q}{\partial \beta_k} = \sum_j a_{kj}\beta_j + \sum_i a_{ik}\beta_i.$$

These are the kth components of $A\boldsymbol{\beta}$ and $A^\mathsf{T}\boldsymbol{\beta}$. Thus the gradient is $(A + A^\mathsf{T})\boldsymbol{\beta}$. If A is symmetric, $A = A^\mathsf{T}$, so this simplifies to $2A\boldsymbol{\beta}$.

Apply these rules to the expanded OLS objective. Since $\boldsymbol{y}^\mathsf{T}\boldsymbol{y}$ is constant, and $X^\mathsf{T}X$ is symmetric,

$$\nabla_{\boldsymbol{\beta}} S(\boldsymbol{\beta}) = -2X^\mathsf{T}\boldsymbol{y} + 2X^\mathsf{T}X\boldsymbol{\beta}.$$

The output is a $p \times 1$ vector, containing one partial derivative for each coefficient.

5.5 Step 5: set the gradient equal to zero

At an unconstrained minimum,

$$-2X^\top \boldsymbol{y} + 2X^\top X\hat{\boldsymbol{\beta}} = \mathbf{0}.$$

Divide by 2 and rearrange:

$$\boxed{X^\top X\hat{\boldsymbol{\beta}} = X^\top \boldsymbol{y}.}$$

These are the *normal equations*. The kth row is precisely the equation obtained by setting $\partial S/\partial \beta_k = 0$ in the scalar derivation.

5.6 Step 6: solve when the inverse exists

If $X^\top X$ is invertible, multiply both sides on the left by its inverse:

$$(X^\top X)^{-1}X^\top X\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \boldsymbol{y},$$

so

$$\boxed{\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \boldsymbol{y}.}$$

The inverse formula is valid only if $X^\top X$ is invertible. The rest of the guide explains exactly when that condition holds.

6 Inequalities and definiteness: the vocabulary

The symbols below express different kinds of comparisons.

Symbol	Meaning
$a > b$	a is strictly greater than b.
$a \geq b$	a is greater than or equal to b. Equality is allowed.
$\boldsymbol{a} = \boldsymbol{b}$	Vectors are equal when all corresponding entries are equal.
$\boldsymbol{a}^\top \boldsymbol{b} = 0$	The vectors are orthogonal (perpendicular in Euclidean geometry).
$A \succeq 0$	Symmetric matrix A is positive semidefinite: $\boldsymbol{v}^\top A\boldsymbol{v} \geq 0$ for all $\boldsymbol{v}$.
$A \succ 0$	Symmetric matrix A is positive definite: $\boldsymbol{v}^\top A\boldsymbol{v} > 0$ for every nonzero $\boldsymbol{v}$.

6.1 Positive semidefinite versus positive definite

A symmetric matrix A is *positive semidefinite* if

$$\boldsymbol{v}^\top A\boldsymbol{v} \geq 0 \quad \text{for every } \boldsymbol{v}.$$

It may equal zero for some nonzero vectors. It is *positive definite* if

$$\boldsymbol{v}^\top A\boldsymbol{v} > 0 \quad \text{for every } \boldsymbol{v} \neq \mathbf{0}.$$

The distinction is the strict inequality and the exclusion of the zero vector.

7 Why $X^\top X$ is positive semidefinite

First, $X^\top X$ is symmetric:

$$(X^\top X)^\top = X^\top (X^\top)^\top = X^\top X.$$

Now choose any vector $\boldsymbol{v} \in \mathbb{R}^p$. Then

$$\boldsymbol{v}^\top X^\top X \boldsymbol{v} = (X\boldsymbol{v})^\top (X\boldsymbol{v}) = \|X\boldsymbol{v}\|^2 \geq 0.$$

The final expression is a squared Euclidean length, a sum of squares. It cannot be negative. Therefore

$$\boxed{X^\top X \succeq 0.}$$

Notice that this proves positive semidefiniteness for every X, including rank-deficient matrices.

7.1 When is it positive definite?

For $X^\top X$ to be positive definite, we need

$$\boldsymbol{v}^\top X^\top X \boldsymbol{v} > 0 \quad \text{for all } \boldsymbol{v} \neq \mathbf{0}.$$

Since this expression is $\|X\boldsymbol{v}\|^2$, this is equivalent to

$$X\boldsymbol{v} \neq \mathbf{0} \quad \text{for all } \boldsymbol{v} \neq \mathbf{0}.$$

In words, the only solution of $X\boldsymbol{v} = \mathbf{0}$ must be $\boldsymbol{v} = \mathbf{0}$. This is exactly the condition that the columns of X are linearly independent.

$$\begin{aligned} X \text{ has full column rank} &\iff \text{Null}(X) = \{\mathbf{0}\} \\ &\iff X^\top X \succ 0 \\ &\iff X^\top X \text{ is invertible.} \end{aligned}$$

8 Rank and linear independence

8.1 What does rank mean?

The *rank* of a matrix is the number of linearly independent columns (equivalently, the number of linearly independent rows). If X has p columns, it has *full column rank* when

$$\text{rank}(X) = p.$$

That means no column is redundant: no column can be written as a linear combination of the other columns.

8.2 A visual example of dependent columns

Consider

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}.$$

The second column is twice the first. The columns are linearly dependent, so $\text{rank}(X) = 1 < 2$. For example,

$$\boldsymbol{v} = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \neq \mathbf{0}, \qquad X\boldsymbol{v} = \mathbf{0}.$$

Because $X\boldsymbol{v} = \mathbf{0}$, we also have

$$X^\top X\boldsymbol{v} = X^\top \mathbf{0} = \mathbf{0}.$$

So $X^\top X$ has a nonzero vector in its null space and cannot be invertible.

8.3 Why square matrices and inverses matter

If X is $n \times p$, then $X^\top$ is $p \times n$. Unless $n = p$, X is rectangular, and it does not have an ordinary two-sided matrix inverse. However,

$$X^\top X \in \mathbb{R}^{p\times p}$$

is square. A square matrix is invertible if and only if its null space contains only the zero vector, equivalently if and only if its columns are linearly independent.

A useful identity is

$$\boxed{\operatorname{rank}(X^\top X) = \operatorname{rank}(X).}$$

Thus $X^\top X$ is invertible exactly when its rank is p, which is exactly when X has full column rank.

9 Inverses, bijections, and rank: how it all connects

The previous sections used the words *invertible*, *singular*, *null space*, *rank*, and *positive definite* side by side, and stated facts such as "a square matrix is invertible if and only if its null space contains only the zero vector". This section explains *why* these ideas belong together. We start from the definition of the inverse and end with a picture of what goes wrong when the columns of X are dependent.

The big idea. A square matrix has an inverse exactly when the map $\boldsymbol{v} \mapsto A\boldsymbol{v}$ is a *bijection*: every output comes from exactly one input. For a matrix, the only way this can fail is that some nonzero direction is collapsed to $\mathbf{0}$. A collapsed direction is also exactly what separates positive *semi*definite from positive definite.

9.1 The definition of a matrix inverse

A square matrix $A \in \mathbb{R}^{p\times p}$ is *invertible* if there is a matrix $A^{-1} \in \mathbb{R}^{p\times p}$ such that

$$AA^{-1} = I_p \qquad \text{and} \qquad A^{-1}A = I_p,$$

where I_p is the $p \times p$ identity matrix (ones on the diagonal, zeros elsewhere). A square matrix with no inverse is called *singular*.

The inverse, when it exists, is unique: if $AB = I_p$ and $CA = I_p$, then

$$C = CI_p = C(AB) = (CA)B = I_pB = B.$$

9.2 Matrices as functions

To understand *when* an inverse exists, think of a matrix as a function. A matrix $A \in \mathbb{R}^{m\times p}$ turns an input vector into an output vector:

$$\boldsymbol{v} \longmapsto A\boldsymbol{v}, \qquad \underbrace{\mathbb{R}^p}_{\text{inputs}} \to \underbrace{\mathbb{R}^m}_{\text{outputs}}.$$

The set of allowed inputs, $\mathbb{R}^p$, is the *domain*; the set where outputs live, $\mathbb{R}^m$, is the *codomain*. The number of columns fixes the domain and the number of rows fixes the codomain. For a design matrix, X takes a coefficient vector $\boldsymbol{\beta} \in \mathbb{R}^p$ and returns a prediction vector $X\boldsymbol{\beta} \in \mathbb{R}^n$.

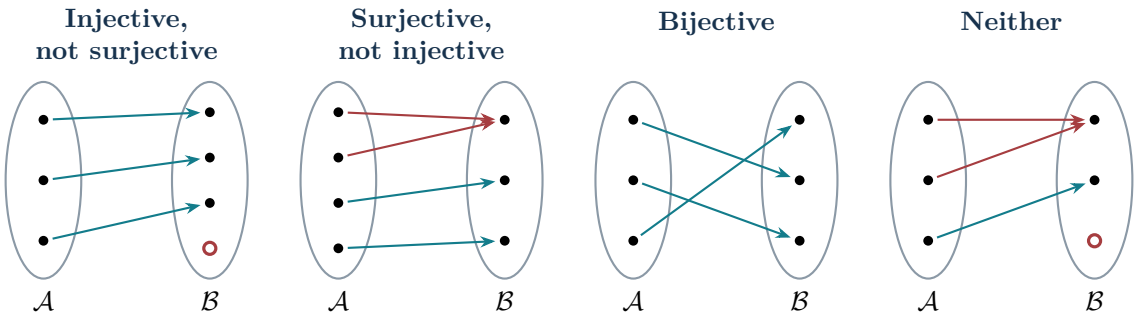
Multiplying matrices is the same as applying one function after another, because $A(B\boldsymbol{v}) = (AB)\boldsymbol{v}$. The identity matrix is the function that does nothing: $I_p\boldsymbol{v} = \boldsymbol{v}$. So the definition of the inverse says:

- $A^{-1}A = I_p$: applying A and then A^{-1} gives back the original input, so A^{-1} *undoes* A;
- $AA^{-1} = I_p$: applying A^{-1} and then A gives back the original vector, so A undoes A^{-1}.

9.3 Injective, surjective, and bijective functions

Let f be a function from a set $\mathcal{A}$ (inputs) to a set $\mathcal{B}$ (codomain).

Property	Definition	In words
Injective (one-to-one)	$f(a) = f(a') \implies a = a'$. Every $b \in \mathcal{B}$ has *at most one* input.	No two inputs collide.
Surjective (onto)	For every $b \in \mathcal{B}$ there is some a with $f(a) = b$. Every b has *at least one* input.	Nothing in the codomain is missed.
Bijective	Both injective and surjective. Every $b \in \mathcal{B}$ has *exactly one* input.	A perfect pairing.



Red arrows: two inputs share an output (injectivity fails). Hollow red circles: outputs that are never reached (surjectivity fails).

Examples with ordinary functions $\mathbb{R} \to \mathbb{R}$.

- $f(x) = x^3$ is bijective: it is strictly increasing, and every b is reached by $x = b^{1/3}$.
- $f(x) = e^x$ is injective but not surjective: no x gives $e^x = -1$.
- $f(x) = x^2$ is neither: $f(-1) = f(1)$, and -1 is never an output.

The same rule can change behaviour if the codomain changes: $x \mapsto x^2$ from $\mathbb{R}$ to $[0, \infty)$ *is* surjective. Injectivity and surjectivity depend on the inputs, the codomain, and the rule together. For matrices, the shape fixes the inputs and the codomain.

9.4 An inverse exists exactly when the map is bijective

A function g is an *inverse* of f if

$$g(f(a)) = a \text{ for every input } a, \qquad \text{and} \qquad f(g(b)) = b \text{ for every } b \in \mathcal{B}.$$

$$\boxed{f \text{ has an inverse} \iff f \text{ is bijective.}}$$

Why? If f is bijective, every b has exactly one input; define $g(b)$ to be that input. Then g undoes f and f undoes g. Conversely, suppose an inverse g exists.

- *f must be injective.* If $f(a) = f(a')$, apply g to both sides: $a = g(f(a)) = g(f(a')) = a'$. Intuitively, if two inputs gave the same output, g could not know which one to return.

- *f must be surjective.* For any b, the input $a = g(b)$ satisfies $f(a) = f(g(b)) = b$. Intuitively, if some b were never an output, $f(g(b)) = b$ could not hold.

Now apply this to matrices. The two conditions $A^{-1}A = I_p$ and $AA^{-1} = I_p$ say precisely that $\boldsymbol{w} \mapsto A^{-1}\boldsymbol{w}$ is an inverse function of $\boldsymbol{v} \mapsto A\boldsymbol{v}$. Conversely, if $\boldsymbol{v} \mapsto A\boldsymbol{v}$ is bijective, its inverse function turns out to be linear, so it can be written as a matrix. Hence:

$$\boxed{A \text{ is invertible} \iff \text{the map } \boldsymbol{v} \mapsto A\boldsymbol{v} \text{ is bijective.}}$$

So the question "does A have an inverse?" is exactly the question "does every output come from exactly one input?".

9.5 Injectivity is decided by the null space

Two sets attached to a matrix $A \in \mathbb{R}^{m\times p}$ answer the two halves of that question:

$$\begin{aligned} \text{Null}(A) &= \{\boldsymbol{v} \in \mathbb{R}^p : A\boldsymbol{v} = \boldsymbol{0}\} \quad \text{(the null space: inputs sent to zero)},\\ \text{Col}(A) &= \{A\boldsymbol{v} : \boldsymbol{v} \in \mathbb{R}^p\} \qquad \text{(the column space: all possible outputs)}. \end{aligned}$$

Since $A\boldsymbol{v} = v_1\boldsymbol{a}_1 + \cdots + v_p\boldsymbol{a}_p$ is a combination of the columns $\boldsymbol{a}_1, \ldots, \boldsymbol{a}_p$ of A, the column space $\text{Col}(A)$ is the set of all linear combinations of the columns. The rank is the number of independent directions in $\text{Col}(A)$: the largest number of linearly independent columns.

For a matrix, injectivity reduces to one question about zero. Because $A\boldsymbol{v} - A\boldsymbol{w} = A(\boldsymbol{v} - \boldsymbol{w})$,

$$A\boldsymbol{v} = A\boldsymbol{w} \iff A(\boldsymbol{v} - \boldsymbol{w}) = \boldsymbol{0} \iff \boldsymbol{v} - \boldsymbol{w} \in \text{Null}(A).$$

Two different inputs collide exactly when their difference is a nonzero vector of the null space. Therefore:

$$\boxed{\begin{aligned} \boldsymbol{v} \mapsto A\boldsymbol{v} \text{ is injective} &\iff \text{Null}(A) = \{\boldsymbol{0}\} \iff A\boldsymbol{v} = \boldsymbol{0} \text{ only for } \boldsymbol{v} = \boldsymbol{0}\\ &\iff \text{the columns of } A \text{ are linearly independent} \iff \text{rank}(A) = p. \end{aligned}}$$

There is more. If $A\boldsymbol{u} = \boldsymbol{0}$ for some $\boldsymbol{u} \neq \boldsymbol{0}$, then

$$A(\boldsymbol{v} + t\boldsymbol{u}) = A\boldsymbol{v} + tA\boldsymbol{u} = A\boldsymbol{v} \qquad \text{for every } t \in \mathbb{R}.$$

A matrix that fails to be injective does not merge just two inputs, as x^2 does: it collapses a *whole line* of inputs onto every output it produces. The part of the input lying along $\boldsymbol{u}$ is destroyed, and nothing can recover it from the output. This is exactly the OLS situation in which $\boldsymbol{\beta}$ and $\boldsymbol{\beta} + t\boldsymbol{v}$ give the same predictions.

9.6 Rank–nullity: why square matrices are special

A standard result of linear algebra, the *rank–nullity theorem*, says that for $A \in \mathbb{R}^{m\times p}$,

$$\boxed{\dim \text{Null}(A) + \text{rank}(A) = p.}$$

Read it as a budget. There are p input directions. Each is either *collapsed* (it lies in the null space) or *survives* into the outputs (it counts towards the rank). Every direction that is collapsed is a direction missing from the outputs.

- **Square matrices** ($m = p$). If nothing is collapsed, the rank is p, so the outputs fill all of $\mathbb{R}^p$: the map is also surjective, hence bijective, hence invertible. If something is collapsed, the rank is less than p, so some outputs are never reached. For square matrices, injective, surjective, and invertible stand or fall *together*, and it is enough to check whether $A\boldsymbol{v} = \boldsymbol{0}$ has a nonzero solution. This is the statement used in the previous section.

- **Tall matrices** ($m > p$, such as a design matrix with $n > p$). The rank is at most $p < n$, so the outputs can never fill $\mathbb{R}^n$. The map $\boldsymbol{\beta} \mapsto X\boldsymbol{\beta}$ is never surjective: most vectors $\boldsymbol{y} \in \mathbb{R}^n$ cannot be produced exactly as $X\boldsymbol{\beta}$. So X never has an inverse, whatever its entries. The best possible case is that X is injective.

This is the real reason OLS works with $X^\mathsf{T}X$. The tall matrix X can never be inverted, but $X^\mathsf{T}X$ is *square* ($p \times p$), so for it "injective" is enough to guarantee "invertible".

9.7 Where semidefiniteness fits in

We showed that $X^\mathsf{T}X \succeq 0$ always, and that positive definiteness can fail only through a direction $\boldsymbol{v} \neq \mathbf{0}$ with $X\boldsymbol{v} = \mathbf{0}$. In fact X and $X^\mathsf{T}X$ collapse *exactly* the same directions:

$$X\boldsymbol{v} = \mathbf{0} \implies X^\mathsf{T}X\boldsymbol{v} = \mathbf{0}, \qquad X^\mathsf{T}X\boldsymbol{v} = \mathbf{0} \implies \boldsymbol{v}^\mathsf{T}X^\mathsf{T}X\boldsymbol{v} = \|X\boldsymbol{v}\|^2 = 0 \implies X\boldsymbol{v} = \mathbf{0}.$$

So

$$\boxed{\operatorname{Null}(X^\mathsf{T}X) = \operatorname{Null}(X).}$$

Since X and $X^\mathsf{T}X$ both have p columns, rank–nullity then gives $\operatorname{rank}(X^\mathsf{T}X) = p - \dim \operatorname{Null}(X^\mathsf{T}X) = p - \dim \operatorname{Null}(X) = \operatorname{rank}(X)$, which proves the identity $\operatorname{rank}(X^\mathsf{T}X) = \operatorname{rank}(X)$ used in the previous section.

The quadratic form $\boldsymbol{v}^\mathsf{T}X^\mathsf{T}X\boldsymbol{v} = \|X\boldsymbol{v}\|^2$ measures how much length the direction $\boldsymbol{v}$ keeps after being mapped by X:

- *positive semidefinite* means no direction has negative length, which is always true;
- *positive definite* means no nonzero direction is squashed to length zero: X collapses nothing.

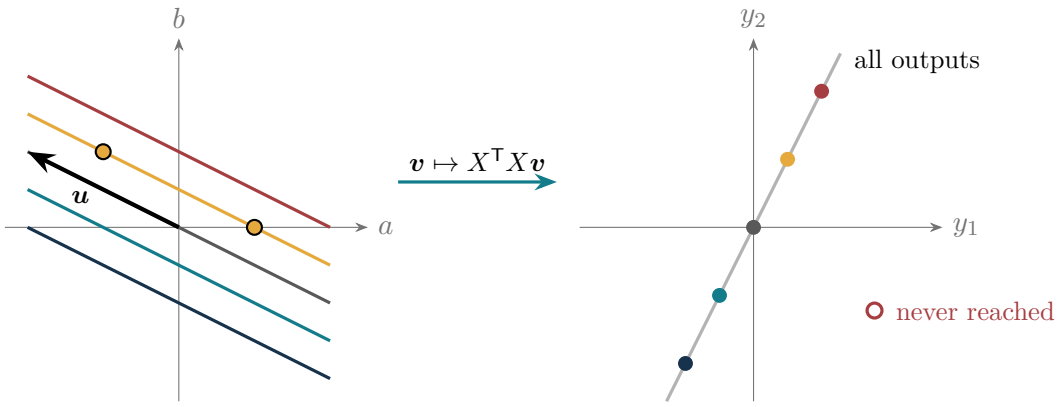
Positive definite = positive semidefinite + nothing collapsed. For the square matrix $X^\mathsf{T}X$, "nothing collapsed" is exactly injectivity, and by rank–nullity injectivity is exactly invertibility.

9.8 A visual picture: why the dependent example is not bijective

Return to the dependent design matrix of the previous section, $X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$. For $\boldsymbol{v} = (a, b)^\mathsf{T}$,

$$X^\mathsf{T}X = \begin{bmatrix} 14 & 28 \\ 28 & 56 \end{bmatrix}, \qquad X^\mathsf{T}X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 14a + 28b \\ 28a + 56b \end{bmatrix} = 14(a + 2b) \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

The output depends on the two numbers a and b only through the *single* number $a + 2b$. Both failures are now visible.



inputs $\mathbb{R}^2$:
coloured lines $a + 2b = c$, $c = -2, \ldots, 2$

outputs $\mathbb{R}^2$
(drawn at reduced scale)

Each coloured line of inputs is parallel to the null direction $\boldsymbol{u} = (-2, 1)^\mathsf{T}$ (black arrow) and lands on the single point of the same colour. The two outlined points $(1, 0)^\mathsf{T}$ and $(-1, 1)^\mathsf{T}$ lie on the same line, so they give the same output. Points off the output line are never reached.

- **Not injective.** Every input on the line $a + 2b = c$ is sent to the same point $14c\,(1, 2)^\mathsf{T}$. For example, $(1, 0)^\mathsf{T}$ and $(-1, 1)^\mathsf{T}$ both give $(14, 28)^\mathsf{T}$. Their difference $(2, -1)^\mathsf{T}$ lies in $\mathrm{Null}(X^\mathsf{T}X) = \mathrm{Null}(X)$, the line through $(-2, 1)^\mathsf{T}$.
- **Not surjective.** Every output is a multiple of $(1, 2)^\mathsf{T}$, so the outputs fill only a line, not the whole plane (rank $= 1$). For example, $X^\mathsf{T}X\boldsymbol{v} = (1, 0)^\mathsf{T}$ has no solution: it would need $14(a + 2b) = 1$ and $28(a + 2b) = 0$ at the same time.
- **The budget.** $2 = 1 + 1$: one input direction is collapsed, so only one survives.

As a result, $X^\mathsf{T}X\boldsymbol{v} = \boldsymbol{b}$ has either *no* solution or *infinitely many*, and never exactly one. An inverse would have to return "the" solution, so $(X^\mathsf{T}X)^{-1}$ cannot exist.

By contrast, Example A in the worked examples below has $X^\mathsf{T}X = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}$, which collapses nothing: it is a bijection of $\mathbb{R}^2$, and its inverse exists.

9.9 Summary

For a design matrix $X \in \mathbb{R}^{n \times p}$, all of the following say the same thing:

$$\begin{aligned} &\text{the columns of } X \text{ are linearly independent} \iff X\boldsymbol{v} = \boldsymbol{0} \text{ only for } \boldsymbol{v} = \boldsymbol{0} \\ &\iff \boldsymbol{\beta} \mapsto X\boldsymbol{\beta} \text{ is injective} \iff \mathrm{rank}(X) = p \iff \mathrm{Null}(X^\mathsf{T}X) = \{\boldsymbol{0}\} \\ &\iff X^\mathsf{T}X \text{ is bijective, i.e. invertible} \iff X^\mathsf{T}X \succ 0. \end{aligned}$$

When they fail, the cause is always the same: a nonzero direction $\boldsymbol{v}$ with $X\boldsymbol{v} = \boldsymbol{0}$, which X and $X^\mathsf{T}X$ both collapse.

10 What this means for OLS: uniqueness and geometry

The OLS objective can be written

$$S(\boldsymbol{\beta}) = \|\boldsymbol{y} - X\boldsymbol{\beta}\|^2 .$$

It is a quadratic function of $\boldsymbol{\beta}$. Its Hessian matrix is

$$\nabla^2_{\boldsymbol{\beta}} S = 2X^\mathsf{T}X.$$

Because $X^\top X$ is positive semidefinite, the objective is convex. A stationary point is therefore a global minimiser.

- If X has full column rank, $X^\top X$ is positive definite. The objective curves upwards in every nonzero coefficient direction, and the minimiser is unique.
- If X does not have full column rank, there is a nonzero $\boldsymbol{v}$ with $X\boldsymbol{v} = \mathbf{0}$. Moving from $\boldsymbol{\beta}$ to $\boldsymbol{\beta} + t\boldsymbol{v}$ leaves the predictions unchanged:

$$X(\boldsymbol{\beta} + t\boldsymbol{v}) = X\boldsymbol{\beta} + tX\boldsymbol{v} = X\boldsymbol{\beta}.$$

  The coefficients may not be unique, even though the best fitted predictions are unique.

10.1 Eigenvalue intuition

Because $X^\top X$ is symmetric and positive semidefinite, all its eigenvalues are nonnegative. It is invertible if and only if none of those eigenvalues is zero. A zero eigenvalue corresponds to a direction in coefficient space along which the objective has no curvature. Such a direction occurs when the columns of X are dependent.

11 Worked examples

11.1 Example A: full column rank and a unique solution

Take a straight-line model with intercept and slope:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad \boldsymbol{y} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}.$$

Compute

$$X^\top X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 3 & 5 \end{bmatrix}.$$

Its determinant is

$$\det(X^\top X) = 2 \cdot 5 - 3 \cdot 3 = 1 \neq 0.$$

Therefore it is invertible, with

$$(X^\top X)^{-1} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix}.$$

Also,

$$X^\top \boldsymbol{y} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}, \qquad \hat{\boldsymbol{\beta}} = \begin{bmatrix} 5 & -3 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ 8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The fitted line is $\hat{y} = 1 + x$. The two columns of X are independent, so the coefficients are uniquely identified.

11.2 Example B: dependent columns and no inverse

Now take

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

The columns are identical. The Gram matrix is

$$X^\top X = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}, \qquad \det(X^\top X) = 2 \cdot 2 - 2 \cdot 2 = 0.$$

It is singular. If $x = 1$ for both observations, the predictions depend only on $\beta_0 + \beta_1$. For any real t,

$$\begin{bmatrix}\beta_0\\ \beta_1\end{bmatrix} = \begin{bmatrix}1+t\\ 1-t\end{bmatrix} \quad \Longrightarrow \quad \beta_0 + \beta_1 = 2.$$

All such coefficient vectors give the same predictions. The issue is not that no best fit exists; the issue is that the coefficients cannot be uniquely separated.

12 Common questions and corrections

12.1 Does X^T need to be invertible?

No. In the usual regression setting X is rectangular, so neither X nor X^T has an ordinary two-sided inverse. The closed-form OLS formula requires the square matrix $X^\mathsf{T}X$ to be invertible.

12.2 If $X^\mathsf{T}X$ is singular, does OLS fail?

Not necessarily. A least-squares minimiser still exists. The coefficient vector may not be unique. One can use the Moore–Penrose pseudoinverse,

$$\hat{\boldsymbol{\beta}} = X^{+}\boldsymbol{y},$$

which gives a minimum-norm least-squares solution.

12.3 Why not always calculate the inverse directly?

Even when $X^\mathsf{T}X$ is mathematically invertible, it can be numerically ill-conditioned. Forming $X^\mathsf{T}X$ squares the 2-norm condition number of a full-column-rank X. Numerical software therefore often uses QR factorisation or singular value decomposition (SVD) rather than explicitly computing $(X^\mathsf{T}X)^{-1}$.

12.4 What about ridge regression?

Ridge regression minimises

$$\|\boldsymbol{y} - X\boldsymbol{\beta}\|^2 + \lambda\,\|\boldsymbol{\beta}\|^2, \qquad \lambda > 0.$$

Its normal equations contain $X^\mathsf{T}X + \lambda I$. Since

$$\boldsymbol{v}^\mathsf{T}(X^\mathsf{T}X + \lambda I)\boldsymbol{v} = \|X\boldsymbol{v}\|^2 + \lambda\,\|\boldsymbol{v}\|^2 > 0 \quad \text{for every } \boldsymbol{v} \neq \mathbf{0},$$

the matrix is positive definite and invertible, even if X is rank-deficient. (This is the standard ridge penalty without an unpenalised intercept adjustment.)

13 Compact derivative rule sheet

For a column vector $\boldsymbol{\beta}$, constant vector $\boldsymbol{a}$, and constant matrix A of compatible dimensions:

$$\begin{aligned}
\nabla_{\beta} c &= \mathbf{0},\\
\nabla_{\beta}(\boldsymbol{a}^\mathsf{T}\boldsymbol{\beta}) &= \boldsymbol{a},\\
\nabla_{\beta}(\boldsymbol{\beta}^\mathsf{T}\boldsymbol{a}) &= \boldsymbol{a},\\
\nabla_{\beta}(\boldsymbol{\beta}^\mathsf{T}A\boldsymbol{\beta}) &= (A + A^\mathsf{T})\boldsymbol{\beta},\\
A = A^\mathsf{T} \implies \nabla_{\beta}(\boldsymbol{\beta}^\mathsf{T}A\boldsymbol{\beta}) &= 2A\boldsymbol{\beta},\\
\nabla_{\beta}\,\|\boldsymbol{y} - X\boldsymbol{\beta}\|^2 &= -2X^\mathsf{T}\boldsymbol{y} + 2X^\mathsf{T}X\boldsymbol{\beta}.
\end{aligned}$$

14 The full argument in six lines

1. OLS minimises the scalar objective $S(\boldsymbol{\beta}) = \|\boldsymbol{y} - X\boldsymbol{\beta}\|^2$.
2. Differentiating with respect to the vector means taking one partial derivative per coefficient and stacking them into a gradient.
3. Setting the gradient to zero gives the normal equations $X^\top X\hat{\boldsymbol{\beta}} = X^\top \boldsymbol{y}$.
4. For every $\boldsymbol{v}$, $\boldsymbol{v}^\top X^\top X\boldsymbol{v} = \|X\boldsymbol{v}\|^2 \geq 0$, so $X^\top X$ is positive semidefinite.
5. It is positive definite, and hence invertible, exactly when $X\boldsymbol{v} = \mathbf{0}$ implies $\boldsymbol{v} = \mathbf{0}$, i.e. when X has full column rank.
6. Under that condition, the unique solution is $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1}X^\top \boldsymbol{y}$.

Matrix notation is not a different calculus.
It is component-wise algebra and calculus written compactly.